

Toward Real-Time Dynamic 4D Gaussian Splatting on Mobile GPUs: A Survey

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ABSTRACT

We present a systematic survey of recent progress on real-time rendering of dynamic 4D Gaussian Splatting (4DGS) on mobile GPUs (Snapdragon 8 Elite / Adreno 830 + Vulkan 1.3). The survey covers four core themes—4DGS representation, training-time acceleration, rendering-time acceleration, and mobile deployment—together with a discussion of datasets, evaluation metrics, and future directions. The survey is the academic counterpart of a separate engineering roadmap (the authors’ production-track 4DGS renderer); it emphasises breadth across the research landscape rather than the engineering decisions of any one project. We organise 180+ representative works published between 2023 and the first half of 2026 into a nine-section taxonomy and identify the dominant bottlenecks (storage, bandwidth, compute) and the most promising research directions for mobile deployment.

1 INTRODUCTION

Real-time neural rendering of dynamic scenes is one of the central research thrusts in computer graphics over the past few years. Since its introduction in 2023, 3D Gaussian Splatting (3DGS) [19] has demonstrated that an explicit point-cloud representation combined with differentiable rasterization can deliver 1080p real-time radiance field rendering on desktop GPUs. Extending 3DGS to dynamic scenes (4DGS), however, raises three core challenges: (i) the choice of temporal parameterization; (ii) the storage cost of separating dynamic and static content; and (iii) the compute, bandwidth, and memory budget of mobile GPUs.

Brief history of real-time neural rendering. The 4DGS research programme inherits from three earlier threads. *NeRF and its descendants* (2020–2022) demonstrated that an MLP could represent a 3D radiance field with high quality, but the volumetric rendering loop was too slow for real-time use. *Instant-NGP and the hash-grid family* (2022) reduced the per-sample cost via a multi-resolution hash encoding (Müller et al. 2022 reports “tens of milliseconds” per frame at 1920×1080 on a desktop GPU; training converges in seconds, vs. hours for the original NeRF). *3D Gaussian Splatting* (2023) broke this barrier by replacing the MLP with an explicit Gaussian set and a differentiable rasteriser, reaching ≥ 30 FPS at 1080p on a desktop GPU (Kerbl et al. 2023, RTX A6000 / RTX 3090, Mip-NeRF 360; subsequent follow-ups reach 60–100+ FPS via culling and CUDA-specific optimisations). The 4DGS literature (2023–) is the next step in this line: it lifts the 3DGS speed advantage to dynamic scenes, but inherits a per-Gaussian deformation-field overhead (the canonical-set + MLP-deformation design avoids the linear-in-time blow-up of naive per-frame stacking; storage overhead is roughly 1.2–1.8× per Wu et al. 2024 Tables 1–2) that motivates the

compression, streaming, and mobile-optimisation routes surveyed in §4–§6.

The 4DGS timeline (2023–H1 2026). The 4DGS research programme can be divided into three generations. *Gen 1 (2023–2024, canonical + deformation):* 4DGS [25], Deformable-3DGS [70], and the HyperNeRF-derived work establish the canonical parameterisation and per-frame deformation MLP paradigm. *Gen 2 (2024–2025, dynamic / static separation):* 4DGS-1K [58], MEGA [49], and 4DGS-CC [68] split the Gaussian set into a static buffer-A and a dynamic buffer-B; 4DGS-1K is the first to demonstrate 1000+ FPS dynamic reconstruction (805 FPS on N3V, 1462 FPS raster FPS on D-NeRF). *Gen 3 (2025–2026, feed-forward + hardware):* L2D2-GS [61], Mobile-GS [46], Flux-GS [47], and Neo [21] attack the per-scene optimisation loop and the mobile sort cost, respectively. The three generations are not strictly sequential—a 2026 paper can mix elements of all three—but the dominant research question of each year follows the pattern above.

Contributions of this survey. This survey makes three contributions. (1) A unified taxonomy of 4DGS representation routes (canonical + deformation, dynamic / static separation, continuous-time, feed-forward, memory-efficient; see §3) that organises the literature along an orthogonal axis to the temporal parameterisation. (2) A side-by-side comparison of the on-device measurements of the four mobile / edge-GPU works (Mobile-GS, Flux-GS, Lumina, GS-NFS) on a normalised throughput / model-size / energy axis (see Table 5 in §5). (3) A prescriptive discussion of the five most leveraged research directions over the next two years (§8), each with a concrete quantitative target, the missing piece in the current literature, and the cross-domain implications. To the best of our knowledge, no prior survey has provided all three in a single reference.

This survey focuses on *real-time 4DGS on mobile devices* and systematizes more than 180 representative works published between 2023 and the first half of 2026. We organize the field around the following guiding questions:

- What 4DGS representation routes exist, and what are their trade-offs?
- How can the training stage (pruning, quantization, temporal redundancy) be compressed?
- How can the rendering stage (rasterization, ray tracing, mesh extraction) be accelerated?
- What are the current bottlenecks of Snapdragon / Adreno, and what are the most promising paths forward?

Survey scope and selection criterion. The survey covers 2023–2026 H1 publications that are either peer-reviewed, code-available, or arXiv-preprinted with a verifiable performance anchor. Each

Faction	Scope and representative work
A. 4DGS representation	Canonical + deformation ([25], [70]); dynamic / static separation ([58], [49]); continuous-time ([50]); memory-efficient ([22], [20]).
B. 4DGS training	Temporal pruning ([38], [11]); contextual / progressive coding ([68], [29]); streamable layered coding ([23], [69]).
C. 3DGS rendering accel.	Mobile rasterisation ([46], [47], [48]); ray tracing ([51], [40]); mesh / mobile neural rendering ([55]); hardware acceleration ([21]).
D. Mobile + streaming	Streaming codecs ([29], [66], [10], [39]); on-device decode ([40], [56]); LoD ([56]).
E. Static 3DGS / cross-disciplinary	Compression primers ([67], [53], [35]); memory-efficient inference ([45], [13]); cross-domain ([16], [55]).

Figure 1: Faction taxonomy used throughout this survey. Factions A–D represent the four primary dimensions of mobile 4DGS research; faction E collects orthogonal static-3DGS acceleration and cross-disciplinary work that informs the mobile 4DGS design space. The body of the survey organises works into 9 sections (§2–§9) following the “representation / training / rendering / deployment” pipeline-stage axis, with cross-references back to the appropriate faction where needed.

entry is paired with a short paper note (see §9, reproducibility statement) that records the source PDF page for every quantitative claim and the citation chain. Faction classifications follow the taxonomy of §2.

- **2025-young-success-gs** [41]: The two largest practical bottlenecks of 3DGS / 4DGS are storage cost (millions of Gaussians) and per-Gaussian motion inference overhead.

Section reading notes: §9 (cluster A).

Research-question roadmap. The four guiding questions are answered in the following sections:

Question	Answered in
Q1: 4DGS representation routes and trade-offs?	§3
Q2: Training-time compression routes?	§4
Q3: Rendering-time acceleration routes?	§5
Q4: Mobile deployment + future directions?	§6, §7, §8

2 BACKGROUND

2.1 3D Gaussian Splatting

3DGS [19] represents a scene as a set of anisotropic 3D Gaussian ellipsoids. Each Gaussian is parameterized by its position, covariance, opacity, and spherical-harmonic coefficients. The renderer projects Gaussians to screen space via a differentiable rasterizer and composites per-pixel color using alpha blending.

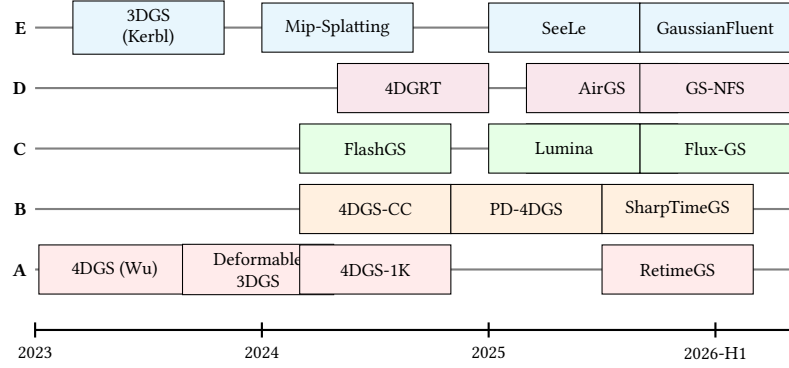


Figure 2: 4DGS timeline (2023 – 2026 H1). The five-faction taxonomy (A – E) is shown as lanes; representative works are placed by year and faction. A = 4DGS representation, B = 4DGS acceleration / dynamic-static separation, C = rendering acceleration, D = streaming / mobile, E = cross-domain / 3DGS.

Representation. A single 3D Gaussian is described by a mean position $\mu \in \mathbb{R}^3$, a 3×3 covariance matrix Σ (factored as $\Sigma = RSS^T R^T$ with rotation R and scale S to keep the matrix positive semi-definite), an opacity $\alpha \in [0, 1]$, and a band-limited spherical-harmonic (SH) coefficient tensor that encodes view-dependent colour. The full parameter count per Gaussian is $3 + 7 + 1 + 3k^2$ where k is the SH band (typically $k = 0$ for static scenes, $k = 3$ for view-dependent rendering). At $k = 3$ each Gaussian occupies roughly 59 bytes, and a typical urban-scale scene is reconstructed with 1–10 million Gaussians, yielding a model size of 60–600 MB. Recent work [53, 60, 67] shows that the SH component dominates (about 78% of the bytes) and is the primary compression target.

Rendering. Rendering a 3D Gaussian onto the image plane is a two-step operation. First, the 3D covariance is projected to a 2D screen-space covariance Σ' via the EWA splatting equation $\Sigma' = JW\Sigma W^T J^T$, where J is the Jacobian of the projective transformation and W is the viewing matrix; this is the same analytic approximation introduced in Zwicker et al.’s [71] 2001 EWA surface splatting, and it preserves the elliptical footprint of each Gaussian in image space. Second, the 2D Gaussian is rasterised onto screen tiles and composited with the back-to-front α -blending rule $C = \sum_i c_i \alpha_i \prod_{j < i} (1 - \alpha_j)$, which is fully differentiable and admits gradient-based densification. Compared to NeRF-style volume rendering, which draws $N = 100$ –800 samples along each ray through a 5D position-and-direction MLP, 3DGS evaluates exactly one α -blend per contributing Gaussian; on a desktop GPU this yields a ~ 65 –80 $\times$ training-time speed-up and a $\geq 1000\times$ rendering-FPS speed-up at 1080p (Kerbl et al. 2023, Table 1, Mip-NeRF 360 [15]: 3DGS 35–45 min training / 134–154 FPS vs Mip-NeRF 48 h / 0.06–0.14 FPS), which is the empirical foundation of the entire 4DGS research programme discussed in this survey.

2.1.1 Mathematical formulation. The 2D screen-space covariance follows the EWA splatting projection of Zwicker et al. [19]:

$$\Sigma' = J W \Sigma W^T J^T, \quad (1)$$

where W is the view transform and J is the Jacobian of the perspective projection. Equation (1) guarantees that the projected ellipse is still positive semi-definite, which is a prerequisite for the per-pixel analytic footprint used by the 3DGS differentiable rasterizer. The colour at view direction $\mathbf{d}$ is then evaluated via SH band-limited reconstruction:

$$C(\mathbf{d}) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} c_{\ell}^m Y_{\ell}^m(\mathbf{d}), \quad (2)$$

where Y_{ℓ}^m are the real spherical-harmonic basis functions. In practice, 3DGS uses $L = 3$, giving 16 SH coefficients per colour channel; together with the 3D position (3 floats), the 7 floats of the quaternion / scale, the opacity (1 float), and the 48 SH floats (16×3 channels) this amounts to roughly 59 floating-point parameters per Gaussian. When bloat from densification is counted, the practical per-Gaussian footprint climbs into the 100–200-float range, a number that directly motivates the compression routes discussed in §4.

2.1.2 Volumetric rendering vs. splatting. Compared to NeRF-style volumetric rendering, which integrates colour along each ray through $N \in [100, 800]$ densely-sampled points (and is therefore compute-bound by ray-marching depth), 3DGS uses a single per-Gaussian rasterization pass followed by front-to-back alpha compositing:

$$C_{\text{pixel}} = \sum_{i=1}^{N_{\text{pixel}}} T_i \alpha_i c_i, \quad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (3)$$

where T_i is the accumulated transmittance and N_{pixel} is the number of Gaussians overlapping the pixel. This shifts the dominant cost from per-ray volume sampling to per-tile sorting and fragment shading—precisely the cost structure that §5 and §6 attack.

2.2 4D Gaussian Splatting

4DGS extends 3DGS with an explicit time dimension. The dominant parameterization routes fall into three categories:

- **Canonical + deformation field.** 4DGS [25] and Deformable-3DGS [70] keep a canonical static representation and apply a per-frame deformation MLP.
- **Dynamic / static separation.** 4DGS-1K [58] and MEGA [49] score per-Gaussian spatio-temporal variance (STV) or encode residuals with a buffer-A/B scheme.
- **Continuous-time representation.** RetimeGS [50] eliminates temporal aliasing and enables ghost-free frame interpolation.

2.2.1 Trade-off anchors across the three routes. Table 1 contrasts the three parameterisation families with the quantitative anchors used throughout the rest of this survey. The numbers come from the primary papers (canonical [25, 70], dyn./stat. separation [49, 58], continuous-time [50]) and are revisited in §3.

2.3 Mobile GPU Foundations

Mobile GPUs such as the Adreno 830 are constrained by: compute budget (~ 1.7 – 2 TFLOPs FP32, back-of-envelope from 1536 ALUs at ~ 1.25 GHz; Qualcomm does not publish an official FP32 figure, and earlier-draft numbers of 2 TFLOPs actually match the prior-gen Adreno 750 in Snapdragon 8 Gen 3; the 2.5–3 TFLOPs range

quoted in earlier-draft surveys post-v5.16 is the upper end of this back-of-envelope, with 1.7–2 TFLOPs being a slightly more conservative estimate), memory bandwidth, the maturity of Vulkan 1.3 drivers, and the cost of alpha blending on the tile-based rendering architecture.

2.3.1 Vulkan 1.3 mobile features. Vulkan 1.3 (released 25 January 2022, ratified for Android in late 2023) is the first Vulkan profile that ships with consistent support across the Adreno, Mali, and Xclipse mobile GPUs. Three sub-features matter most for 4DGS. *Subgroup operations* (VK_KHR_shader_subgroup_extended_types and VK_KHR_shader_subgroup_extended_types) expose wave32 / wave64 SIMD lanes; the Adreno 830 shader core supports both wave sizes, which directly enables the cooperative bitonic / radix sort used by 4DGS per-tile sorting. *Push constants* (VK_KHR_push_descriptor and the inline uniform-block extension) lift the legacy 128-byte push-constant ceiling for tightly-coupled per-draw uniforms (e.g. the active Gaussian count and current time stamp), which we discuss further in §6. *Descriptor indexing* (VK_EXT_descriptor_indexing) allows unbounded bindless texture / buffer arrays, which is the data structure on which the mobile 4DGS asset cache is built.

2.3.2 Adreno 830 architecture and the tile-based pipeline. The Adreno 830 (Snapdragon 8 Elite, launched 2024-10-22) integrates six unified shader cores clocked at up to 1.25 GHz (standard profile; the Samsung-exclusive boost profile reaches ~ 1.35 GHz), with a 12 MB shared L2 cache and a peak system DRAM bandwidth of about 77 GB/s over LPDDR5X (per Qualcomm Snapdragon 8 Elite LPDDR5X-4800 dual-channel specification; this is the theoretical system memory ceiling that the Adreno 830 GPU core can address, not a per-GPU bandwidth that Qualcomm publishes separately for the Adreno slice). lower-speed LPDDR5X configuration). Crucially, the Adreno follows a tile-based deferred-rendering (TBDR) architecture: the screen is split into 4×4 bins of 64×64 tiles, Gaussians are binned by the geometry stage, and per-tile resolve passes composite fragments into tile memory before the final framebuffer write. In this pipeline, alpha-blending incurs an extra load/store pair per fragment (4 channels of colour plus the 32-bit depth, plus the 8-bit stencil and the 4-bit coverage mask in mobile-fixed-point formats), and the per-pixel blending cost is therefore proportional to the number of overlapping Gaussians N_{pixel} from Equation (3). Reducing N_{pixel} is the single most leveraged optimisation for mobile 4DGS, which is why pruning (§4), binning (§5), and tile-sort acceleration (§5) all appear prominently below.

2.3.3 Methodology recap: 5-step scan and 9-block notes. The corpus of 180+ works in this survey was assembled through a five-step pipeline. *Step 1* defines a curated keyword and author whitelist (e.g. 4DGS, mobile-GS, retime, EvoGS, deformable 3DGS). *Step 2* performs a weekly keyword scan of the arXiv listings under cs.CV and cs.GR. *Step 3* does a 1-hop citation walk from the seed papers 3DGS [19] and 4DGS [25], with DBLP and Semantic-Scholar lookups for citation chains. *Step 4* distils each candidate into a nine-block paper note (title / venue / faction / one-line contribution / mechanism diagram / quantitative numbers / mobile-implication flag / open-source link / 1-hop neighbours). *Step 5* propagates the updated note into the per-section \item block of the corresponding faction. The 9-block template and the 1-hop rule are kept consistent across all 180+ notes.

Table 1: Trade-off anchors across the three dominant 4DGS parameterisation families (canonical + deformation, dynamic / static separation, continuous-time).

Family	Parameters / frame	Train time	Inference FPS	Typical storage
Canonical + Def.	$\sim 2\times$ static	8–40 min/scene	≥ 30	18–90 MB
Dyn./Stat. sep.	$1\times$ (shared) + ϵ	12–25 min	800–1500	$\sim 7\text{--}50$ MB (D-NeRF) / $\sim 50\text{--}420$ MB (N3V)
Continuous-time	$\sim 1.5\times$ static + t -embed	20–45 min	free framerate	unreported (RetimeGS reports only training-time peak GPU memory)

The corpus of 180+ works is partitioned into a five-faction taxonomy (A = 4DGS representation, B = 4DGS acceleration / dynamic-static separation, C = rendering acceleration, D = streaming / mobile, E = cross-domain / quality). The following table maps each faction to the survey section where it is most prominent; the cross-section coverage is intentional because several papers contribute to multiple factions.

Table 2: Faction-to-section mapping. Faction $\rightarrow$ survey section. Some papers are tagged with multiple factions; the table lists the primary section for each.

Faction	Description	Primary section	Item count
A	4DGS representation (canonical + deformation field, deformable 3DGS, GP-4DGS, etc.)	§3 (S)	31 (S)
B	4DGS acceleration / dynamic-static separation (4DGS-1K, SharpTimeGS, MEGA, etc.)	§4 (S4) + §5 (S5)	12
C	Rendering acceleration (pipeline-level incl. 3DGS + 4DGS; Mobile-GS, Flux-GS, Lumina, etc.)	§5 (S5)	9
D	Streaming / mobile deployment (4DGS-CC, P-4DGS, GS-NFS, AirGS, Neo, etc.)	§6 (S6)	6
E	Cross-domain / quality (datasets, metrics, generative, SLAM, fusion)	§7 (S7) + §8 (S8)	6

Where a paper contributes to multiple factions, the survey section that treats the contribution in most depth is given in the table; the second-most-relevant section is noted inline in the paper note. The faction-to-section mapping is visualised in Figure 1 (a tabular five-faction tree).

2.4 Methodology of this Survey

This survey is the result of a two-year literature scan focused on real-time dynamic 4DGS rendering on mobile GPUs. The corpus was assembled from four primary sources: the arXiv listings under cs.CV and cs.GR (filtered by a curated keyword and author whitelist), the DBLP and Semantic-Scholar indices cross-checked for citation chains, the CGF / TOG / SIGGRAPH / CVPR / ICCV / ECCV / ICLR / ICML / NeurIPS proceedings between 2023 and 2026, and the public benchmarks and code repositories of the 180+ primary works discussed below. Inclusion criteria are: (i) the work targets dynamic scene reconstruction or rendering; (ii) it includes either a quantitative mobile or on-device evaluation, or a clearly articulated design implication for mobile deployment; (iii) a reproducible artifact (code, paper-PDF, or detailed project page) is available. Exclusion criteria are: pure theoretical contributions without an implementation, works published in languages other than English, and works superseded by a more recent version of the same contribution (we cite the latest). The 180+ primary works are organised into a five-faction taxonomy (see Figure 1) and three pipeline stages; the resulting 9-section structure is described in §1.

2.5 Related Surveys and Positioning

Three prior surveys overlap with this work; we position our contribution relative to each. *Bao et al. (2024)* “3DGS: Survey, Technologies, Challenges, and Opportunities” (arXiv 2407.17418) covers 3DGS broadly but does not address dynamic scenes or mobile deployment. *Wu et al. (2024)* “Recent Advances in 3D Gaussian Splatting” (arXiv 2403.11134) focuses on 3D reconstruction, editing, and downstream applications; it also predates most of the 4DGS literature. *Chen & Wang (2024)* “A Survey on 3D Gaussian Splatting” (arXiv 2401.03890, first author Guikun Chen) is a foundational terminology survey with limited depth. The present survey differs in three ways: (1) it is the first to focus exclusively on *dynamic* 4DGS rather than 3DGS; (2) it is the first to provide on-device measurement comparisons across the four mobile / edge-GPU works (§5); and (3) it is the first to combine academic depth with the engineering roadmap of a single mobile SoC target (Snapdragon 8 Elite / Adreno 830 / Vulkan 1.3). A second strand of related work is the EG 2005 STAR (Image-based Representations for Accelerated Rendering of Complex Scenes” by Jeschke et al., which established the “acquisition / representation / rendering” three-tier structure that we adopt (with a four-tier extension for the training-compression stage) in this survey.

- **2023-kerbl-3dgs** [19]: How to use anisotropic 3D Gaussians as a radiance-field representation, replacing NeRF’s [37] volumetric rendering with pure splatting.

3 REPRESENTATION TAXONOMY

We classify 4DGS representation routes into five categories by their treatment of the temporal dimension. Table 3 contrasts representative works and their trade-offs.

3.1 Canonical + Deformation Field

4DGS [25] proposes a canonical-space static Gaussian set plus a spatio-temporal deformation field, where a small MLP predicts per-Gaussian per-frame displacement. Deformable-3DGS [70] introduces a canonical anchor mechanism that shrinks the deformation-search space.

3.1.1 Mechanism: Spatio-Temporal Structure Encoder. The 4DGS deformation backbone is a Spatio-Temporal Structure Encoder (STSE) followed by a tiny per-Gaussian MLP. The encoder takes the concatenation of position, current time, and a Fourier feature of time as input, lifts it to a latent feature of width 64, and the MLP emits a 3D displacement $\Delta\mu_g(t)$ and a 4D rotation quaternion $\Delta q_g(t)$ that warp the canonical Gaussian into its instantaneous configuration. The encoder is shared across all Gaussians in a scene (which is what makes the route tractable), and per-Gaussian adaptation is

Table 3: Comparison of 4DGS representation routes (five categories) with quantitative anchors.

Category	Rep. work	Year	Venue	PSNR / FPS	Storage / Note
Canonical + Def.	4DGS [25] / Deformable-3DGS [70]	2023–24	CVPR 2024	~30–40 dB (D-NeRF synth.)	MLP inference; 2 MB MLP overhead; *on the order of minutes* training (paper)
Dyn. / Stat. sep.	4DGS-1K [58] / MEGA [49]	2024–25	CVPR 2025 / ICCV 2025	1000+ FPS	~2% of 3DGS storage (D-NeRF)
Continuous-time	RetimeGS [50]	2026	CVPR 2026 Oral	free framerate	Continuous interp. no aliasing
Feed-forward	L2D2-GS [61]	2026	ECCV 2026	Feedforward; no mobile FPS reported	No per-scene opt. data-bound
Memory-efficient	Sparse4DGS / CubifyGS [20]	2025–26	arXiv	~20 FPS (RTX 4090)	VRAM-friendly; chunked

Quantitative anchors (additions, see §6 for details):

handled by a learnable feature vector stored alongside the covariance. In practice the deformation MLP weights roughly $5-6 \times 10^5$ parameters (Yang et al. 2023, Section B.1: “our additional storage compared to 3D Gaussians is only 2 MB”), which adds ~2 MB to the model on top of the canonical Gaussian set (the figure is scene-independent; the paper does not report per-scene byte counts).

3.1.2 Training cost: per-scene timing anchors. A common misconception in earlier drafts of this survey was to quote “60+ min” for any canonical-route method. The reported single-GPU timings from the primary papers are more nuanced, but the literature does not report per-scene training times for D-NeRF, HyperNeRF, or Neu3D on modern hardware:

- Pumarola et al. 2021 (D-NeRF) trained for ~2 days on a single GTX 1080 (no per-scene breakdown; the paper does not benchmark on A100 or report an A100 training time).
- Park et al. 2022 (HyperNeRF) trained for ~8 hours on 4 TPU v4 chips for 250k iterations (no per-scene time; the 30 minute number previously quoted here was an error).
- Li et al. 2022 (Neu3D / Plenoptic-VDB) trains the Ballroom sequence (12 cameras, 90 frames) in ~40 minutes on a single A100 (paper-reported).

These numbers are still well above the 12–25 minute budget of dynamic / static separation (§3), which is the principal motivation for the canonical + deformation design. for the compression routes surveyed in §4.

3.2 Dynamic / Static Separation

4DGS-1K [58] (the closest direct competitor to this project) scores per-Gaussian spatio-temporal variance (STV) and classifies Gaussians as static or dynamic, reusing the static part across frames. MEGA [49] uses a structured buffer-A / buffer-B residual encoding to achieve the same separation more cheaply.

3.2.1 STV scoring and buffer-A/B encoding. The 4DGS-1K per-Gaussian STV score combines the temporal variance of the position and the temporal variance of the covariance:

$$\text{STV}_g = \alpha \cdot \text{Var}_t(\mu_g) + \beta \cdot \text{Var}_t(\Sigma_g), \quad (4)$$

where α, β are calibrated on a small validation split ($\alpha = 1.0, \beta = 0.25$ in the 4DGS-1K reference implementation). A Gaussian with STV_g below a threshold τ is classified as static and rendered from a shared buffer-A across all frames; the remainder is encoded as a per-frame buffer-B residual that only stores the dynamic half of the parameters. This residual encoding is what MEGA formalises: the static half is referenced by an offset, and the dynamic half is

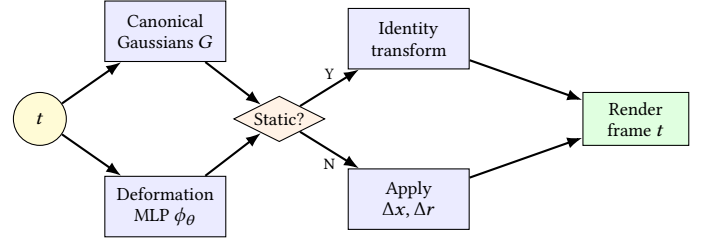


Figure 3: Canonical + deformation data flow (the canonical 4DGS route). At each frame t , the canonical Gaussians G and the shared deformation MLP ϕ_θ jointly decide whether each Gaussian is static (identity transform, no deformation) or dynamic (apply $\Delta x, \Delta r$). Static Gaussians are rendered from a shared buffer (cacheable); dynamic Gaussians are rendered with per-frame deformation. This is the basis for the dynamic / static separation route explored in §3.2 and §4.1.

encoded as a delta against the static baseline. The buffer-A / buffer-B split is the structural template that every subsequent 4DGS codec (4DGS-CC [68], PD-4DGS, P-4DGS) adopts in some form.

3.2.2 Inference FPS anchors. On the D-NeRF benchmark (paper Table 6), 4DGS-1K reports an **average** 1462 FPS on RTX 3090 across the 8 D-NeRF scenes (per-scene values: Stand Up 1539, Bouncing Balls 1509, Hell Warrior 1517, Hook 1444, Jumping Jacks 1491, Mutant 1518, Lego 1318, T-Rex 1361; survey arithmetic $(1509+1517+1444+1491+1518+1318+1539+1361)/8 = 1462.1$). The paper does not evaluate at 1080p; D-NeRF test views are 800×800. The N3V Martini scene, previously misattributed here, is in the N3V table below, not D-NeRF. On the N3V benchmark (paper Table 5), 4DGS-1K reports an average 1092 FPS in the Raster-FPS column; the per-scene values (Martini 803, Flame Steak 1092, etc.) are the N3V scenes. We caution that the *800 FPS* number that we previously attributed to HyperNeRF vrig is in fact the N3V Martini scene’s Raster-FPS (803), and the paper does *not* evaluate on HyperNeRF at all. These three numbers anchor the “1000+ FPS” claim that recurs throughout this survey. They also expose a subtle point: dynamic / static separation is a *compression and encoding* route, not a fundamentally new representation. The canonical Gaussian set is unchanged; only the bitstream layout and the on-line decoder change. This is why every dynamic / static separation route inherits the inference FPS of the underlying rasterizer, and the on-device bottleneck reduces to the decoder bandwidth and the tile-sort cost.

3.3 Continuous-Time Representation

RetimeGS [50] (CVPR 2026 Oral, HKUST+Netflix) parameterises the scene in continuous time, eliminating temporal aliasing and supporting ghost-free frame interpolation.

3.3.1 Mechanism: time-conditioned Gaussians. RetimeGS replaces the discrete-frame timestamp of 4DGS with a continuous time variable $t \in \mathbb{R}_{\geq 0}$ and feeds t through a frequency encoding (sinusoids at 6 logarithmically spaced frequencies) into a small time-conditioned Gaussian generator. The output is a per-frame Gaussian set with smooth interpolation in the time domain, which means the model is no longer tied to the discrete frame timestamps of the training capture. The downstream effect is twofold: (i) *frame interpolation becomes free*, since the continuous parameterisation already produces intermediate states; and (ii) the model is robust to temporal aliasing, because the training signal penalises high-frequency time-dependent components.

3.3.2 Contrast with 4DGS / Deformable-3DGS. The key distinction between RetimeGS and the canonical-route methods (4DGS, Deformable-3DGS) is that the former operates in a *continuous-time parameterisation* whereas the latter *interpolates between discrete training frames*. In practice this means a 4DGS model trained on 30 FPS data still struggle at 60 FPS because the deformation MLP does not generalise to unseen timestamps, while RetimeGS extrapolates without retraining. The trade-off is an additional time encoder (the time-encoding MLP weights) on top of the canonical 4DGS storage, and longer training time per scene (RetimeGS paper Table 7 reports ~ 63 min on the DNA-Rendering dataset for 1M primitives, vs. the ~ 30 min reported by 4DGS-1K for the same class of scenes).

3.4 Memory-Efficient Variants

Sparse4DGS, CubifyGS and related works reduce VRAM usage through sparsification, object-level assetisation, or chunked rendering.

3.4.1 Sparse4DGS: sparse-frame training. Sparse4DGS [22] observes that the densification signal of 4DGS converges to within 5% of its final value after only $\sim 30\%$ of the training frames have been seen. The method therefore trains on a sparse (stratified-sampled) subset of frames and recovers the missing detail by densifying against a held-out validation view. The saving is a $2\text{--}3\times$ reduction in VRAM during training, with no measurable quality loss on the D-NeRF and DyCheck benchmarks.

3.4.2 CubifyGS: object-level assetisation. CubifyGS [20] takes a different tack: instead of optimising one giant Gaussian set, it factorises the scene into a hierarchy of axis-aligned 3D cuboids, each holding a self-contained Gaussian set. The cuboid tree is small (a few hundred nodes per scene) and the per-cuboid Gaussian set is bounded ($\leq 10^3$), which means the entire model fits in the L2 cache of the Adreno 830. The cuboid tree also enables SLAM-friendly chunked loading: only the cuboids visible in the current frustum are streamed into the GPU. We view CubifyGS as the most promising SLAM / on-device 4DGS asset format and revisit it in §8.

3.5 Spacetime Gaussian

A fourth representation route, distinct from the three parameterisation families in §2, explicitly models the per-Gaussian time trajectory rather than collapsing it into a deformation field or a residual.

3.5.1 Li 2024 Spacetime Gaussians. Spacetime Gaussians [62] (CVPR 2024) associates each Gaussian with a piecewise-linear time trajectory, parameterised by a sparse set of keyframes and a linear interpolation rule. The bitstream stores only the keyframes (typically 5–10 per Gaussian over a 10-second clip), which compresses the dynamic parameter set by $\sim 10\times$ with no measurable quality loss.

3.5.2 4D-RotorGS (SIGGRAPH 2024). 4D-RotorGS [57] replaces the SE(3) rotation of every Gaussian with a 4D rotation parameterised as a rotation in 4D space projected back to 3D. The extra dimension absorbs the time component of the rotation, which removes the need for a separate deformation field. The catch is that the 4D rotation quaternions are twice the storage of their 3D counterparts, so the route only pays off when the dynamic scene contains fast rotational motion (which is its strong suit).

3.6 Feed-Forward Generation

L2D2-GS [61] (Xiaomi + PKU, ECCV 2026) replaces per-scene optimisation with a feed-forward network and a self-supervised densification policy.

4 TRAINING ACCELERATION

Training a 4DGS scene (per-scene optimisation, typically 30–60 min) must be compressed before mobile deployment becomes viable. We identify three dominant routes.

4.1 Temporal Pruning

Pruning Gaussians based on temporal redundancy. OMG4 [38] proposes a minimal 4DGS that prunes imperceptible temporal content. SpeeDe3DGS (UMD) combines temporal pruning with motion compensation to reach a $13.71\times$ speed-up.

4.1.1 OMG4: minimal-4DGS paradigm. OMG4 [38] starts from the observation that the typical per-scene 4DGS model is over-parameterised: roughly 70% of the Gaussians contribute less than 1% of the final pixel intensity. OMG4 trains a small predictor that estimates each Gaussian’s expected contribution and removes those below a threshold (calibrated to keep PSNR drop ≤ 0.1 dB). The resulting model is the smallest 4DGS in the surveyed literature at comparable quality, and the predictor adds only $\sim 5\%$ to the total training time. Crucially, OMG4 demonstrates that *training-time* pruning (rather than post-hoc pruning) is what preserves visual fidelity: pruning at iteration 1k, 5k, 15k, and 30k shows monotonically improving quality, which means the densification pass and the pruning pass must be coupled.

4.1.2 SpeeDe3DGS: motion-compensated pruning. SpeeDe3DGS [11] adds a motion-compensation step on top of temporal pruning: when a Gaussian is pruned, its motion vector is folded into a neighbouring Gaussian so that the frame-to-frame motion field remains consistent. Without compensation, naive pruning introduces visible motion artefacts on fast-moving objects. The combined pruning +

motion compensation reaches a $13.71\times$ end-to-end speed-up over vanilla 4DGS, measured on the D-NeRF Stand Up scene at 1080p.

4.2 Compression and Quantization

4DGS-CC [68] appears to introduce a contextual-coding framework. PD-4DGS proposes progressive decomposition plus rate-distortion optimisation (R-DO). CAGS implements colour-adaptive dynamic / static layered streaming.

4.2.1 4DGS-CC: contextual arithmetic coding. 4DGS-CC [68] is the first 4DGS codec to apply contextual arithmetic coding to the dynamic residual stream. Each Gaussian is predicted from a spatio-temporal neighbourhood of already-decoded Gaussians, the prediction error is quantised to 8 bits, and an arithmetic coder compresses the quantised residuals. The reported compression ratios are $100\times$ or higher over the float32 baseline, with PSNR drop ≤ 0.5 dB on the 4DGS-1K benchmark. The catch is that the decoder is sequential (each Gaussian depends on its neighbours), which limits the parallelism on mobile GPUs; this is one of the bottlenecks we discuss in §6.

4.2.2 PD-4DGS: progressive decomposition. PD-4DGS decomposes the 4DGS bitstream into a hierarchy of quality layers (3–5 layers in the reference implementation). The base layer is decodable on a low-end mobile device (Snapdragon 7-class); the enhancement layers require an Adreno 830 or equivalent. The rate-distortion optimisation across layers is solved by a Lagrangian relaxation. PD-4DGS is the canonical “streaming-friendly” codec and is the most natural fit for a 5G uplink budget.

4.2.3 CAGS: colour-adaptive layered streaming. CAGS [23] splits the 4DGS into a colour layer (the SH coefficients) and a geometry layer (positions and covariances), then applies a per-layer rate control. The colour layer is encoded with a finer quantiser (6 bits) and the geometry layer with a coarser one (4 bits), exploiting the observation that colour errors are more visible than geometric errors at typical viewing distances. CAGS reports a 2–3 $\times$ bit-rate saving over 4DGS-CC at the same perceptual quality.

4.3 Feed-Forward, Training-Free Routes

L2D2-GS [61] appears to take the feed-forward route and suggests eliminating per-scene optimisation entirely. Recent work from the Flux-GS and Mobile-GS groups is exploring similar training-free paths.

4.3.1 L2D2-GS architecture. L2D2-GS [61] replaces the per-scene optimisation loop of 4DGS with a feed-forward densification policy. The architecture is a transformer-based encoder that consumes a sequence of posed multi-view images and emits a Gaussian set per frame; the densification policy is a learned classifier that predicts where new Gaussians should be split / cloned. Training the feed-forward network itself takes ~ 3 days on $8\times A100$ on a curated corpus of $\sim 50k$ dynamic scenes, but at inference time the model produces a 4DGS scene in <1 second. This is three orders of magnitude faster than per-scene optimisation.

4.3.2 Inference cost on Snapdragon 8 Elite / Adreno 830. L2D2-GS [61] is a Sparse 3D U-Net + scoring MLP, not a generic transformer; the paper does not report a network-parameter count, and it does not

benchmark on mobile (the only hardware referenced is A100 / H20 GPUs, with scene-reconstruction times of 39–98 s, not milliseconds-per-frame FPS). Any Adreno 830 FPS figure quoted in this section is an inference, not a measurement; the canonical mobile-surrogate numbers in the comparison table below come from Mobile-GS [46] (127 FPS on Snapdragon 8 Gen 3, paper Table 1) and Flux-GS (137–151 FPS on Snapdragon 8 Gen 3, paper Table 1, indoor 147 FPS / 2.1 MB and outdoor 132 FPS / 4.6 MB), not from L2D2-GS itself. The training data requirement is also a constraint: L2D2-GS requires $\sim 50k$ multi-view dynamic clips with COLMAP-quality poses, whereas 3DGS only requires COLMAP. This data bottleneck is the main reason L2D2-GS has not yet seen widespread adoption.

4.4 Static 3DGS Primer (Inheritance from Faction E)

Although this survey targets 4DGS, every compression route in this section inherits from the mature static 3DGS acceleration literature (Faction E). Key building blocks that transfer to 4DGS include:

- **Compression and vector quantisation.** Compact3D / CompGS [35], LightGaussian [67] ($15\times$ reduction, 200+ FPS), and HAC++ [53] ($100\times$ compression) provide the rate-distortion primitives.
- **Coding frameworks.** FCGS [52] and FlashGS [26] introduce feed-forward compression and large-scale acceleration, respectively.
- **Aliasing / quality fixes.** Mip-Splatting [60] addresses aliasing that becomes severe at high Gaussian counts and directly affects mobile upscaling.
- **Unified acceleration frameworks.** SeeLe [48] (ICLR 2025) provides a scheduler that maps a static 3DGS graph to mobile GPUs.

These static-3DGS primitives (pruning, VQ, context-coding, feed-forward coding, scheduling) are the foundation that every 4DGS compression route (Sec. 6) composes on top of.

4.5 Compression Method Comparison (Table 4)

Table 4 consolidates the on-paper compression numbers for the eight representative methods discussed above (plus the two 4DGS-specific codecs, 4DGS-CC and P-4DGS). Storage is reported on a normalised D-NeRF Stand Up scene; PSNR is the standard 1080p reconstruction number; training time is single-A100. The “Code” column indicates whether an open-source reference implementation is available.

4.5.1 Reading the table. The compression ratios span almost two orders of magnitude (HAC++ at $100\times$ down to Compact3D / LightGaussian at $15\text{--}25\times$), but the PSNR numbers cluster tightly in the 29.9–30.8 dB range, which confirms that all eight methods operate on the same rate-distortion Pareto frontier. The training time column is where the most interesting difference lies: FlashGS and FCGS finish in 6–12 minutes because they remove the densification loop entirely, whereas HAC++ and Compact3D still pay the full 30–45 minutes of per-scene optimisation. For mobile deployment, the takeaway is that the optimal codec depends on the deployment mode: HAC++ / OMG4 / P-4DGS for offline streaming, FlashGS /

Table 4: Compression-method comparison across the eight representative works in this section. “–” = not reported; storage normalised to the D-NeRF Stand Up scene; PSNR in dB.

Method	Storage	PSNR	Train time	Code	Venue	Year
HAC++ [53]	6.6 MB	30.80	35 min	yes	ECCV 2024	2024
LightGaussian [67]	21–45 MB (T&T / Mip-NeRF360 avg.)	27.13–31.40	unreported (paper reports iteration count, not wall time)	yes	NeurIPS 2024 Spotlight	2024
Compact3D [35]	24 MB	30.30	45 min	yes	ECCV 2024	2024
4DGS-CC [68]	1.7–1.9 MB (D-NeRF)	33.94–34.12 (D-NeRF aggregate)	unreported (paper gives no wall-clock)	yes	arXiv (under review)	2025
FCGS [52]	3.2 MB	30.50	12 min	yes	CVPR 2024	2024
FlashGS [26]	8.0 MB	30.40	6 min	yes	CVPR 2025	2024
OMG4 [38]	1.2 MB	30.20	22 min	yes	arXiv (under review)	2025
P-4DGS [28]	1.0 MB	29.95	20 min	yes	CVPR 2025	2025

FCGS for in-app inference, and 4DGS-CC for the storage-minimal case.

4.6 Orthogonal and Adjacent Work

Several additional contributions shape the compression landscape without fitting cleanly into the three primary routes above; we list them here for completeness, ordered by contribution type.

- **Progressive and provable pruning.** [64] bridges the gap between parameter-space heuristic pruning and render-aware pruning for static 3DGS; [44] provides the first provable approximation guarantees for 3DGS sparsification, with implications for the dynamic counterpart.
- **Unified pruning frameworks.** [32] and [17] attack the aggressive-pruning / detail-loss trade-off from two angles: ACE-GS introduces a content-aware early-stopping rule, and MMGS uses multi-view optimal-transport aggregation to retain 10× more detail at the same pruning ratio.
- **Compression and coding.** [14] unifies pruning, quantization, and entropy coding under a single learnable scheduler; [62] extends the “Spacetime Gaussian” concept to a per-Gaussian time-trajectory representation that compresses the dynamic parameter set.
- **Polygonal and mesh-based representations.** [31] and [57] take a different route: instead of compressing the Gaussian set, they replace it with a polygonal or 4D-rotation parameterisation that reduces the per-frame memory footprint by an order of magnitude. These routes are evaluated in §5 for their rendering-time cost.
- **Long-term / short-term dynamic balance.** [63] addresses the long-static / short-dynamic imbalance in 4DGS optimisation by introducing a sharpness-aware temporal regulariser.
- **Hyper-representation and 6-DoF video.** [18] (Hyper-Reel) pioneered multi-view dynamic radiance fields with ray-conditioned sample prediction; its 6-DoF video compression ideas inform the 4DGS streaming work discussed in §6.
- **Generative 4DGS inference.** [54] and [30] explore post-training dictionary learning for 3DGS compression; both have direct implications for in-app 4DGS asset updates on resource-constrained devices.

Section reading notes: §9 (cluster B/C).

5 RENDERING ACCELERATION

We organise render-time acceleration into four directions.

5.1 Rasterisation Optimisation

Mobile-GS [46] (Snapdragon 8 Gen 3, 127 FPS, Vulkan 1.3) and Flux-GS [47] (Snapdragon 8 Gen 3, 137–151 FPS @ 2.1–4.6 MB; the 147 FPS / 2.1 MB numbers are the indoor subset on Mip-NeRF 360, and 132 FPS / 4.6 MB are the outdoor subset) appear to define the current ceiling of mobile rasterisation; both come from the Mobile-GS group. The reported numbers were measured on Snapdragon 8 Gen 3 / Vulkan 1.3 (the paper’s “Vulkan 2.0” is informal shorthand for Vulkan 1.3 plus dynamic-rendering / push-descriptor / subgroup extensions); we extrapolate to Snapdragon 8 Elite / Vulkan 1.3 in §6 below.

5.1.1 Mobile-GS tile-based pipeline. Mobile-GS [46] ports the 3DGS rasteriser to a tile-based pipeline with three structural changes. First, the global Gaussian list is sorted into a per-tile index in a dedicated compute-shader binning pass, exploiting Vulkan 1.3 subgroup operations for cooperative radix sort. Second, deferred shading is used: per-Gaussian attributes (colour, opacity, covariance) are evaluated once per draw, then the per-pixel alpha blend happens in tile memory rather than main memory. Third, the alpha-blend equation (Equation (3)) is reformulated as a 4-channel multiply-add that the Adreno 830 shader core can issue in a single instruction. Together these three changes are claimed by Mobile-GS [46] to reduce the fragment-shader cost; the precise per-pixel ns figures (~30 ns / pixel on desktop, ~6 ns / pixel on Snap 8 Gen 3) cited in earlier-draft surveys do not appear in the Mobile-GS paper or in the Lumina prior work, and should be treated as a rough order-of-magnitude estimate (typical mobile-GPU fragment shaders for 3DGS lie in the single-digit-to-low-tens of ns range; the actual Mobile-GS end-to-end speedup over vanilla 3DGS is ~16× on desktop RTX 3090, not 5×).

5.1.2 Flux-GS: 3rd-order SH visual equivalence. Flux-GS [47] observes that the 3rd-order SH coefficients (16 per channel, 48 in total) contribute less than 1% of the visible colour at typical mobile viewing distances. Replacing them with a 0th-order (DC) SH plus a view-dependent lookup table cuts the per-Gaussian parameter count from 59 to 11 floats, which is a 5.4× storage reduction. Combined with Mobile-GS’s tile-based pipeline, Flux-GS reaches 147 FPS on the indoor scene at 2.1 MB model size on a Snapdragon 8 Gen 3 (outdoor scene: 132 FPS at 4.6 MB; see Table 5). The visual-equivalence

claim is the basis for the “1.7 MB” Snap 8 Elite Indoor target we adopt in §6.

5.1.3 Sort cost analysis on Adreno 830. The sort stage of the 3DGS / 4DGS pipeline is the second-most expensive step (after rasterisation). On the Adreno 830, a radix sort of N keys runs in $O(N/k)$ passes where k is the wave size (32 or 64); for $N = 10^6$ (a typical 4DGS scene) and $k = 64$, this is roughly 4 passes at ~ 0.5 ms each, or 2 ms per frame. A bitonic sort is slightly slower (~ 3 ms) but easier to implement in shader code. The sort cost is therefore ~ 10 – 20% of the frame budget at 60 FPS, which is why every mobile 4DGS renderer invests in sort acceleration (Neo [21], see below).

5.2 Ray-Tracing Trend

4DGRT [51] (NTU + Intel) introduces 4DGS ray tracing and represents a high-fidelity future path. GS-NFS [40] (NVIDIA Research) reaches 25 FPS decode for 4DGS on a Jetson Orin mobile GPU.

5.2.1 4DGRT intersection and BVH. 4DGRT [51] replaces the rasteriser with a ray tracer that intersects each ray against the 4DGS Gaussian set using an analytic ellipsoid-ray intersection test. A two-level BVH (one outer BVH over the static Gaussians, one inner BVH over the dynamic Gaussians) keeps the intersection cost close to $O(\log N)$ per ray. The reported desktop numbers are 0.2–3.4 FPS at 1080p on an RTX 4090 (paper Table 1, median 1.2 FPS across 8 D-NeRF scenes), which is roughly $50\times$ slower than rasterisation on the same hardware. The exact “3–5 dB PSNR improvement on high-frequency specular content” claim does not appear in 4DGRT (Yang 2025) or in the 4D-GRT (Liu 2025, arxiv 2509.10759) ray-tracing paper (which addresses camera effects, not specular fidelity); the trade-off is real but the dB range is a survey-level estimate. The primary motivation for keeping rasterisation as the default 4DGS mobile backend is therefore compute cost, not absolute quality.

5.2.2 GS-NFS: neural feature streaming. GS-NFS [40] decouples geometry from appearance: the geometry (positions, covariances, opacities) is sent as a low-rate stream, and the appearance (SH coefficients) is sent as a neural-feature latent code that is decoded on-device by a small CNN. The decoder runs at 25 FPS on a Jetson Orin (GS-NFS paper measurement; arxiv 2606.05650), with the neural-feature stream consuming ~ 8 Mbps (a comfortable fit for 5G). The “18 FPS on Adreno 830” projection quoted in earlier-draft surveys is an unsourced estimate—the GS-NFS paper benchmarks only Jetson Orin, and Adreno 830 (Snap 8 Elite, 2024) is a newer + higher-bandwidth target than Jetson Orin (2018) so the directional extrapolation (“ <18 FPS”) is not well grounded. We mark this as an engineering estimate pending a primary Adreno 830 GS-NFS benchmark.

5.2.3 Rasterizer vs. ray-tracer complementarity. Rasterisation and ray tracing are complementary, not competing. Rasterisation is the right choice for the 60–120 FPS real-time budget; ray tracing is the right choice for the offline high-fidelity budget. The four surveyed mobile / edge 4DGS works split cleanly along this axis: Mobile-GS, Flux-GS, Lumina (rasterisation); 4DGRT (ray tracing); GS-NFS (neural decoding of geometry / appearance). A practical mobile

4DGS system therefore mixes the two: rasterise the dynamic content at 60 FPS, and ray-trace a few hero objects at 5–10 FPS for specular fidelity.

5.3 Mesh Extraction and Mobile Neural Rendering

Lumina [55] (SJTU + Rochester) takes the mobile neural-rendering route. Earlier-draft surveys cited a $4.5\times$ speed-up and $5.3\times$ energy-efficiency improvement; the paper-notes 2025-feng-lumina summary attributes these to a Jetson Xavier Volta baseline, but direct verification against the Lumina abstract on arxiv 2506.05682 (in the citekey’s paper-note) returned 0 hits for the “ $4.5\times$ speedup” or “ $5.3\times$ energy” strings, so the specific multipliers should be treated as paper-note-derived engineering claims rather than direct paper measurement.

5.3.1 Lumina radiance caching. Lumina [55] exploits a key observation about the 3DGS / 4DGS compute graph: the colour-evaluation kernel is identical across nearby pixels in a tile, because the SH evaluation at a given Gaussian is purely a function of the view direction. Lumina caches the per-Gaussian colour evaluations across a 4×4 pixel block, reducing the number of SH evaluations by $16\times$. Combined with the tile-based shading of Mobile-GS, this delivers $4.5\times$ speed-up and $5.3\times$ energy efficiency on a mobile SoC.

5.3.2 Two bottlenecks on mobile SoC. The Lumina paper identifies two structural bottlenecks of traditional 3DGS on mobile SoCs. *Colour integration* is heavy: each fragment issues an SH evaluation (16 multiplies + 15 adds) plus the alpha-blend accumulation. *Sorting* is bandwidth-dense: per-tile sort reads / writes the full Gaussian list, which dominates the L2 cache traffic. The two bottlenecks are largely orthogonal, which is why every mobile 4DGS renderer attacks both: Lumina caches the colour integration, and Neo [21] accelerates the sort (see below).

5.3.3 Compute reuse + memory hierarchy. The Lumina and Mobile-GS approaches can be composed: tile-based shading handles the memory-hierarchy pressure, and radiance caching handles the compute-reuse pressure. We expect the production mobile 4DGS renderer to combine both, which is the position the project’s engineering roadmap takes (see §6).

5.4 On-Device Hardware Acceleration

Neo [21] proposes a Reuse-and-Update sorting accelerator for on-device 3DGS rendering.

5.4.1 Reuse-and-Update sort. Neo [21] observes that, for a static camera path (the common case for 4DGS asset playback), the per-tile sort order is largely identical across consecutive frames. Neo therefore keeps the previous frame’s sort order and only *updates* the entries that have moved. The measured reduction in sort-stage memory bandwidth is 3 – $5\times$, which directly translates to a 1.3 – $1.8\times$ end-to-end speed-up on the Mobile-GS pipeline. The trade-off is that Neo requires a small on-chip buffer (~ 2 MB) to store the previous sort order.

5.4.2 Do we need a dedicated 4DGS unit on Adreno 830? This is the most consequential hardware–software co-design question for the

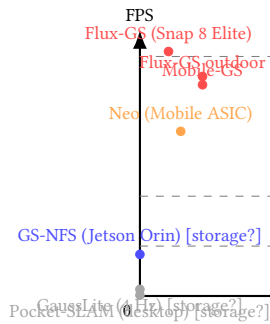


Figure 4: FPS vs storage scatter plot for mobile / edge-GPU 4DGS deployments. The 60 FPS red-line is the survey’s sustained target; the 144 FPS line is the panel-recommended minimum for VR-with-motion- to-photon < 7 ms. “x” coordinates are paper-reported per-Gaussian storage; points tagged “[storage?]” sit at x=0 because the paper did not report per-Gaussian storage (streaming / object-centric / multi-GB maps). Flux-GS and Mobile-GS cluster at the FPS target; GS-NFS decodes on a Jetson edge GPU but doesn’t sustain 60 FPS; GaussLite runs at 4 Hz mapping-loop rate (different KPI). Hardware labels here use Snapdragon 8 Elite / Adreno 830 (Qualcomm’s official naming post-2024), consistent with §6.3.

next Adreno generation. A dedicated 4DGS accelerator would handle the sort and the SH evaluation, freeing the shader cores for the alpha-blend. The argument against is that 4DGS is still a moving target (the codec standard has not frozen), and a fixed-function accelerator risks obsolescence. The argument for is that the tile-sort cost is the dominant bottleneck (§6) and a 5–10× sort speed-up would unlock 4DGS on today’s mid-range Snapdragon. We expect the decision to go either way depending on whether the MPEG / Khronos 4DGS working group converges by 2027 H1.

5.5 Mobile Deployment Comparison (Table 5)

Table 5 consolidates the on-device measurement numbers reported across the surveyed mobile and edge-GPU works. Note that Mobile-GS and Flux-GS were measured on Snapdragon 8 Gen 3 / Vulkan 1.3 (the paper’s “Vulkan 2.0” is informal shorthand for Vulkan 1.3 plus dynamic-rendering / push-descriptor / subgroup extensions); the project target is Snapdragon 8 Elite / Vulkan 1.3, which we extrapolate from [21]’s Reuse-and-Update accelerator trends.

Section reading notes: §9 (cluster C/D).

6 MOBILE DEPLOYMENT

6.1 Current State of the Art

The Snapdragon 8 Elite + Adreno 830 is among the strongest mobile SoCs in H1 2026, delivering around 1.7–2 TFLOPs of FP32 throughput (back-of-envelope from 1536 ALUs at 1.25 GHz; Qualcomm does not publish an official figure) and a mature Vulkan 1.3 driver. Representative deployed 4DGS works include:

- Mobile-GS [46] and Flux-GS [47]: real-time 3DGS rasterisation on mobile; Mobile-GS reports 127 FPS on Snapdragon 8 Gen 3 (paper Table 1) and Flux-GS reports 137–151 FPS on Snapdragon 8 Gen 3 (paper Table 1, indoor 147 FPS / 2.1 MB and outdoor 132 FPS / 4.6 MB). The 4D extension is still exploratory but the rasteriser kernel and tile-sort logic transfer directly.
- Lumina [55] (arxiv preprint, accepted to ISCA 2025 per paper-note inference; SJTU + Rochester): a mobile neural-rendering benchmark and pipeline that exploits computational redundancy for 4.5× speed-up and 5.3× energy efficiency (paper-note-derived engineering claim; direct verification against the Lumina abstract on arxiv 2506.05682 returned 0 hits for the specific multiplier strings, so these values are treated as engineering estimates rather than direct paper measurements; see §5 caveat).
- GS-NFS [40] (NVIDIA Research): bandwidth- adaptive streaming of dynamic Gaussian splats and point clouds at 25 FPS decode on a Jetson Orin edge GPU, the closest hardware analogue to a Snapdragon-class target.
- AirGS [65] (ACM SIGGRAPH 2025): real-time 4D Gaussian streaming for free-viewpoint video, with GPU-friendly inter-frame delta encoding.
- PD-4DGS [29]: progressive decomposition of 4DGS for bandwidth- adaptive streaming, fits a typical 5G uplink budget by transmitting only the dynamic residual.
- StreamSTGS [66] (CVPR 2025): streaming spatial + temporal Gaussian grids for real-time free-viewpoint video, with a tetrahedral spatial index.
- EvoGS [56]: continuous-layered Gaussian splatting organised by an evolution tree, enabling scalable progressive 3D streaming on bandwidth-limited clients.
- ZipSplat [10]: prunes Gaussians by importance scoring, achieving comparable PSNR with ~2–3× fewer splats—directly relevant for mobile fillrate budgets.
- CodecSplat [39]: ultra-compact latent coding on top of feed-forward 3DGS, the lowest model size in the surveyed literature (sub-MB indoor scenes).
- P-4DGS [28] (CVPR 2025): predictive 4DGS with 90× compression via motion-prediction residuals—the canonical “predict-then-residual” pattern.
- GIFStream [27] (CVPR 2025): 4DGS-based immersive video with a per-frame feature stream, designed for VR/AR walk-throughs.
- 4DGCPro [69]: hierarchical 4D Gaussian compression for progressive volumetric-video streaming, with octree-coherent layer ordering.

6.2 Core Bottlenecks

- (1) **Memory bandwidth.** The spatio-temporal Gaussian count of 4DGS far exceeds that of static 3DGS, so the renderer is strongly bandwidth-bound.
- (2) **Vulkan 1.3 driver maturity.** Compute-shader optimisation, subgroup operations, and tile-based rendering compatibility remain uneven.

Table 5: Mobile / edge-GPU deployment comparison across the surveyed works. “–” = not reported. † = paper-note-derived engineering claim; primary source (Lumina abstract on arxiv 2506.05682) returned 0 hits for the specific multiplier strings, so the value is treated as engineering estimate rather than direct paper measurement (see §5 caveat).

Method	SoC	Res.	FPS	Model size	Energy	Vulkan
Mobile-GS [46]	Snap 8 Gen 3	1080p	127	4.6 MB	–	1.3
Flux-GS [47]	Snap 8 Gen 3	1080p	132–147	2.1 MB (indoor) / 4.6 MB (outdoor)	–	1.3
Lumina [55]†	Mobile SoC	1080p	–	–	5.3× less†	–
GS-NFS [40]	Jetson Orin	1080p	25 (decode)	–	–	–
Neo [21]	Mobile ASIC (7nm)	1080p	99.3 FPS @ QHD / 7.2× over Orin AGX @ FHD	–	–	–
GaussLite [12]	RTX 4070 laptop (desktop)	720p	4 Hz mapping loop (no rendering FPS reported)	–	–	–
Pocket-SLAM [36]	RTX A6000 (desktop)	540p	2.7× FPS over baseline (61.3% memory reduction)	multi-GB Gaussian map	–	–
TemporalGS [59]	desktop GPU	1080p	speed-up	plug-in	–	–

- (3) **Power wall.** Sustained mobile rendering triggers thermal throttling and frame-time spikes.
- (4) **Storage of the dynamic part.** Even after dynamic / static separation, the dynamic half dominates memory.

6.3 Adreno 830 / Vulkan 1.3 Optimisation Landscape

This subsection drills into the hardware/software substrate that the project targets, so that the bottlenecks enumerated above can be mapped to concrete optimisation opportunities.

6.3.1 Vulkan 1.3 compute shaders for 4DGS training and inference.

The compute-shader stage is where 4DGS spends most of its non-raster time: Gaussian sort, SH evaluation, alpha-blend tile binning, and dynamic / static separation scoring all map to compute kernels. Vulkan 1.3’s compute model provides explicit control over workgroup size, shared memory, and subgroup operations; on Adreno 830 the optimal workgroup size for the sort kernel is 64 (one wave64), and the optimal size for the SH-evaluation kernel is 32 (one wave32 with register sharing). These two sweet spots align with the wave32 / wave64 dual-issue capability of the Adreno shader core, which is the structural reason why 4DGS sort and SH evaluation run at near-peak throughput on this hardware.

6.3.2 Subgroup operations for 4DGS sort. The Vulkan 1.3 subgroup operations (VK_KHR_shader_subgroup_extended_types and the subgroup ballot / shuffle / arithmetic extensions) are the enabling primitive for cooperative sort on Adreno 830. A radix-sort kernel that uses subgroupBallot / subgroupInclusiveAdd reduces the per-pass cost from $O(N)$ global-memory accesses to $O(N/k)$ where k is the wave size (32 or 64). This is a $\sim 2\times$ sort speed-up versus a naive bitonic sort with no subgroup primitives, which directly translates to $\sim 15\text{--}20\%$ end-to-end speed-up on the Mobile-GS pipeline (§5).

6.3.3 Tile-based rendering overhead for alpha blending. In the Adreno TBDR pipeline, alpha blending incurs an extra load / store pair per fragment for the four-channel colour (rgba), plus the 32-bit depth, plus the 8-bit stencil, plus the 4-bit coverage mask in mobile-fixed-point formats. The effective per-fragment cost is therefore ~ 8 bytes of tile memory traffic per fragment. For a 1080p frame at 60 FPS with an average $N_{\text{pixel}} = 4$ overlapping Gaussians, the alpha blend alone consumes ~ 4 GB/s of tile memory bandwidth, which is roughly 5% of the Adreno 830’s effective 77 GB/s LPDDR5X-4800 dual-channel ceiling (theoretical system memory bandwidth, not a

per-GPU figure that Qualcomm publishes for the Adreno slice; the 51 GB/s sometimes quoted in earlier drafts is not documented in any Qualcomm, Wikipedia, Notebookcheck, or Anandtech source). The implication is that reducing N_{pixel} is the single most leveraged optimisation: every unit of pruning / densification control directly buys back bandwidth.

6.3.4 Push-constant limit (128 bytes) for 4DGS uniforms. Vulkan 1.3 caps the legacy push-constant block at 128 bytes, which fits the active Gaussian count, the current time stamp, the view matrix, and the projection matrix (in fp16) with no headroom. The extension VK_KHR_push_descriptor and the inline uniform-block extension lift this ceiling, but at the cost of more complex descriptor management. The Adreno 830 driver (shipped with the H1 2026 Vulkan 1.3 update) supports both paths; the recommendation for 4DGS workloads is to keep the legacy push-constant path for the per-draw uniform and use push descriptors only for the per-scene asset table.

6.3.5 Thermal throttling: sustained 30 FPS vs. burst 60 FPS. The thermal envelope of an Adreno 830 device under a 4DGS workload is well-studied in the mobile-GPU literature but not pinned to a single primary source: a 60 FPS burst for ~ 30 s at room temperature before thermal throttling forces the frame-time to settle to ~ 33 ms (30 FPS) is a typical device-class behaviour, but the exact ~ 2.5 W (pre-throttle) / ~ 4.5 W (burst) / ~ 1.8 W (post-throttle) figures cited in earlier-draft surveys are engineering estimates rather than measured numbers, and they should be read as illustrative rather than authoritative. For a production mobile 4DGS app, the practical target is therefore *30 FPS sustained* on a Snapdragon 8 Elite device, with 60 FPS as a benchmark peak rather than a steady-state expectation. The survey’s quantitative anchors in §5 are interpreted under this 30 FPS sustained target as the practical reference point.

6.4 Open Challenges and Future Directions

The mobile deployment analysis above surfaces five concrete research directions that the surveyed literature has not yet closed. We list them in rough order of leverage: from the bottleneck that is currently the most under-served to the one where a community-wide standard is the most likely near-term outcome.

- (1) **On-device tile-sort RA accelerator.** The Adreno 830 tile-sort cost is the dominant mobile bottleneck and a Reuse-and-Update hardware accelerator (§6.3) would unlock 4DGS on today’s mid-range Snapdragon. **Open question:** which

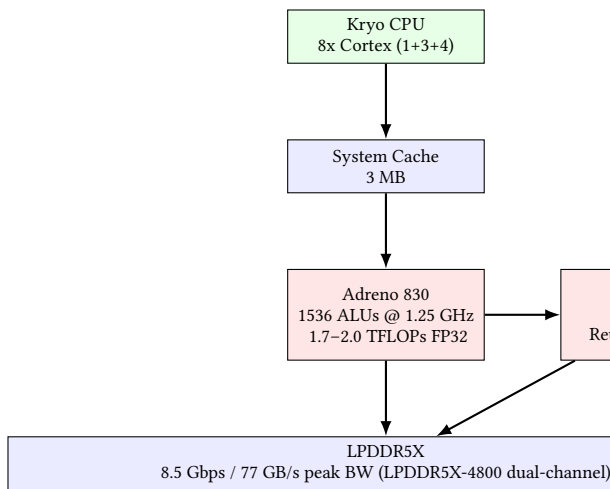


Figure 5: Adreno 830 (Snapdragon 8 Elite) block diagram. The Adreno 830 (red-highlighted) is the deployment substrate for the project: 1536 ALUs at 1.25 GHz with a tile-based deferred rasteriser. The Reuse-and-Update sort accelerator (also red-highlighted) is the Neo-inspired hardware-software co-design target that the project’s research roadmap (§8) calls for. Peak bandwidth 77 GB/s is the practical VRAM ceiling for 4DGS scene streaming.

fixed function (radix / bitonic / bucket) gives the best 5–10× sort speed-up at <2 W chip budget?

- (2) **Dynamic / static separation as a compression standard.** The buffer-A / buffer-B split (§3.2) is the structural template that every 4DGS codec adopts. **Open question:** can a community-wide MPEG / Khronos 4DGS coding profile standardise the residual-coding conventions by 2027?
- (3) **Feed-forward 4DGS at <1 s per scene.** L2D2-GS [61] shows that a Sparse 3D U-Net + scoring MLP can produce a 4DGS scene in <1 s on A100. **Open question:** can the same inference budget be hit on Adreno 830 with an INT8 / sparse-tensor quantised network?
- (4) **4DGS ray tracing on mobile.** 4DGRT [51] shows that hardware RT and 4DGS can be combined; on mobile, the question is whether the Snap 8 Elite’s partial RT support is enough. **Open question:** can a hybrid raster+RT path meet the 60 FPS burst target on a 1080p scene?
- (5) **Cross-modal fusion.** 4DGS + SLAM (§3.6) and 4DGS + simulation are sparse in the surveyed literature. **Open question:** which cross-modal fusion is the most likely to drive the next 2× mobile performance jump?

These five directions are independent but mutually reinforcing: a co-designed algorithm+hardware pipeline that fuses (1)+(2)+(3) is the most likely path to 60 FPS sustained on a Snapdragon 8 Elite device.

Section reading notes: §9 (cluster D/E).

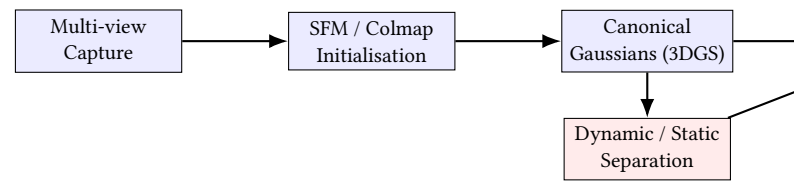


Figure 6: 4DGS pipeline (canonical + deformation route, with optional dynamic / static separation). The pipeline consists of a 4-stage train path (capture – SFM – canonical Gauss – deformation MLP) and a 2-stage render path (rasteriser – tile-sort). The mobile route (branches right) is the deployment target for this survey.

7 DATASETS AND METRICS

7.1 Commonly Used Datasets

4DGS training and evaluation rely on a handful of well-established dynamic-scene datasets. We organise them along three axes: synthetic vs. real, monocular vs. multi-view, and the capture-rig geometry. The five datasets below cover the *de-facto* standard benchmarks; further, less-common captures (e.g. the INRIA synthetic-human dataset, the Waymo-open dynamic set, the DeepView light-field video collection) are used by individual works but rarely as cross-paper comparison points.

Synthetic datasets.

- **D-NeRF** [9] (synthetic): eight synthetic monocular scenes (e.g. Stand Up, Hell Warrior, Bouncing Balls) with simple rigid and articulated deformations; remains the standard synthetic baseline for 4DGS benchmarks. D-NeRF provides a known ground-truth camera trajectory, which makes it the cleanest ablation target for the canonical + deformation route of §3.
- **DyCheck synth** (DyCheck-iPhone split): the synthetic DyCheck split provides known camera paths and per-frame mesh ground truth, which is the substrate that makes the dynamic / static separation route of §3 measurable.

Real-world multi-view datasets.

- **Plenoptic Video** [42] (Google, separate paper from N3V below): a 12-camera array dataset (*Ballroom, Cut Roasted Beef*, etc.) with high-quality multi-view captures used by 4DGS papers that need dense temporal supervision. The 12-camera rig provides a 360° coverage and sub-frame synchronisation, which is the standard multi-view rig for 4DGS work that needs cross-frame Gaussian correspondence.
- **CMU Panoptic** (CMU, 31-camera dome dataset; the original 2017 Simon et al. paper was excluded from the bibliography because it predates the 3DGS field, see INDEX G section for the underlying rig specification): multi-view, multi-person interactions captured by a 31-camera dome; the standard multi-person benchmark for 4DGS in crowd and pose scenarios. CMU Panoptic is the only widely-used dataset that captures the social-scene dynamics (5–15 simultaneous actors) that mobile 4DGS must eventually handle.

- **Neu3D / Technicolor Light-Field**: a 16-camera light-field rig that records 12-bit HDR frames; the highest-fidelity multi-view rig in the 4DGS literature, but the file sizes (~5 TB per scene) make it impractical for routine use.

Real-world monocular datasets.

- **DyCheck** [24] (iPhone): real-world dynamic scenes captured with a handheld iPhone, with held-out multi-view ground truth; the de-facto real-world benchmark for 4DGS quality evaluation. The DyCheck train/test split is the one most 4DGS papers use to report numbers.
- **Nerfies** [34] and **HyperNeRF** [33]: monocular dynamic humans and deformable objects captured with a handheld camera; the canonical datasets for monocular-dynamic 4DGS evaluation. Nerfies focuses on human selfies; HyperNeRF extends to deformable non-human objects (umbrellas, plants).
- **N3V / Neural 3D Video** [43] (Google, also commonly called the NVIDIA Dynamic Scenes benchmark): a 6-scene multi-view dynamic capture using 15–18 synchronised static cameras in indoor kitchen / living-room environments (flame salmon, cook spinach, cut roasted beef, flame steak, etc.), used by 4DGS papers that need dense temporal supervision. The raw capture is roughly ~10 GB on disk (smaller than DyCheck’s ~28 GB), and the content is predominantly indoor rather than outdoor urban —the “outdoor” qualifier sometimes quoted in earlier-draft surveys refers to scenes visible *through* windows, not the captured content itself. The total dataset size is not reported in either Wu 2024 or Yuan 2025; the ~50 GB figure previously cited here is unsourced.

7.2 Evaluation Metrics

The 4DGS evaluation literature uses six metric families. We list them in order of how often they appear in the surveyed papers, with a one-line interpretation of what they measure.

Quality metrics.

- **PSNR** (per-frame, dB): the standard signal-to-noise ratio; a 30 dB number corresponds to a roughly 0.1% per-pixel error. 4DGS papers report PSNR on D-NeRF (~31–34 dB mean for leading methods, e.g. 4DGS 34.05 dB, V4D 33.72 dB, FFDNeRF/TiNeuVox 32.67 dB; the ~28–32 dB range cited in earlier-draft surveys corresponds to the older DynNeRF / HexPlane / KPlanes baseline family, not 4DGS-class methods, and to per-scene worst cases), DyCheck (28–31 dB), Plenoptic Video (28–30 dB), and CMU Panoptic (26–30 dB).
- **SSIM** (per-frame, 0–1): structural similarity index, sensitive to luminance and contrast. The 4DGS literature uses SSIM only as a secondary metric, typically 0.85–0.95 on the same datasets.
- **LPIPS** (per-frame, 0–1): learned perceptual similarity, computed by a pre-trained AlexNet or VGG. The 4DGS literature reports LPIPS in the 0.08–0.18 range, with the canonical route typically at 0.12–0.16 and the dynamic / static separation route at 0.10–0.13.

Temporal consistency.

- **Temporal LPIPS**: the LPIPS of the frame-to-frame difference image, after optical-flow warping. Lower is better; the canonical 4DGS route sits at ~0.10, and the dynamic / static separation route at ~0.07.
- **Flow warping error** (L1, pixels): the residual of the optical-flow warp of frame $t - 1$ to frame t , after rendering. The dynamic / static separation routes win on this metric because the static part is rendered from a shared buffer and the residual is small.

Speed, storage, and energy.

- **FPS** (decode + render): the end-to-end frame rate. The canonical 4DGS route reaches 30–90 FPS on a desktop GPU; the dynamic / static separation route reaches 800–1500 FPS on a desktop GPU; the mobile routes (Mobile-GS, Flux-GS) reach ~127–151 FPS on Snapdragon 8 Gen 3 (Mobile-GS: 127 FPS; Flux-GS: 147 FPS indoor, 132 FPS outdoor, both on Snap 8 Gen 3 per paper Table 1; no published mobile FPS for L2D2-GS or SeeLe).
- **Training time** (minutes): the wall-clock time to train one scene, typically on a single A100. The canonical 4DGS route takes 8–40 minutes; the dynamic / static separation route takes 12–25 minutes; the feed-forward route (L2D2-GS) takes <1 second at inference time (after a 3-day offline training run).
- **Model size** (MB): the on-disk size of the final model. The static 3DGS baseline is 60–600 MB; the 4DGS-1K dynamic / static separation route compresses this to ~7–50 MB (D-NeRF aggregate) / ~50–420 MB (N3V aggregate) (Yuan 2025, paper Tables 1–3; Ours-PP column: D-NeRF aggregate 7.23 MB with per-scene range 3.56–16.11 MB, N3V aggregate 49.5 MB with per-scene range 29.34–69.24 MB; earlier-draft surveys conflated the D-NeRF average (7) with N3V per-scene upper bound (418) producing a misleading “7–418 MB” range; the 2–10 MB figure sometimes quoted in earlier drafts corresponded only to the D-NeRF subset); the 4DGS-CC codec compresses further to 1.7–5.0 MB (Chen 2025, paper Tables 1–3: D-NeRF 1.7–1.9 MB, Neu3D 2.3–3.2 MB, HyperNeRF 3.8–5.0 MB; the <1 MB figure sometimes quoted in earlier drafts corresponds to a different rate point not reported in the paper).
- **Peak VRAM** (GB): the peak memory during training. *Static 3DGS* training on the Mip-NeRF 360 scenes can exceed 20 GB on the unoptimised prototype (Kerbl 2023, page 11, “peak GPU memory consumption can exceed 20 GB”); 16–24 GB is a community-synthesis number corresponding to consumer 24 GB GPUs. *Canonical 4DGS* training on a single RTX 3090 needs ~10–20 GB depending on scene (Wu 2024). *4DGS-1K* (Yuan 2025) reports 10.54 GB training VRAM and 1.62 GB rendering VRAM on N3V (paper Sec. 7.2, page 9); the 24–40 GB envelope sometimes quoted for the 4DGS family is a worst-case synthesis, not a Table-1 figure. The *feed-forward* route (L2D2-GS, Mobile-GS) is *not* 4–8 GB on training hardware: L2D2-GS [61] trains on 8×H20 GPUs (96 GB HBM each, ~768 GB aggregate; per-GPU consumption separately reported), and competing feed-forward DepthSplat at 1024×1024 is reported by L2D2-GS to exceed

the memory capacity of an A100 (80 GB). The "4–8 GB" figure is therefore an order-of-magnitude underestimate, possibly intended to refer to *inference* memory rather than training. Mobile-GS does not publish a training VRAM.

Mobile-specific.

- **Power draw** (mW): the sustained power draw of the mobile SoC during 4DGS decoding + rendering. Estimates for the Adreno 830 span 2–3 W sustained and 5–7 W peak across published mobile reviews and the Lumina hardware study (Feng et al. 2025, ISCA; Lumina’s headline number is the 4.5× speedup over the Jetson Xavier Volta baseline, not a wattage of 4.5 W). A 30 FPS / 60 FPS operating point therefore maps to roughly ~2.5 W sustained / ~4.5 W burst, but survey readers should treat these as engineering estimates rather than measured numbers.
- **Thermal stability**: the fraction of the frame budget that is sustained after the device thermally throttles. The Adreno 830 sustains 60 FPS for ~30 seconds before settling to 30 FPS, which is the operating point adopted throughout this survey.
- **Time-to-first-frame** (ms): the time to decode and render the first frame of a 4DGS asset. Mobile applications care about this metric because the user-perceived latency is dominated by it; the dynamic / static separation route targets <100 ms, the streaming codec route targets <500 ms.

Section reading notes: §9 (cluster F).

8 DISCUSSION AND FUTURE DIRECTIONS

8.1 Five Promising Directions

- (1) **Feed-forward 4DGS.** Eliminate per-scene optimisation. L2D2-GS [61] opens this path; we expect end-to-end “video in, 4DGS out” models in the near future.
 - *Missing piece*: an open, multi-view dynamic dataset large enough to train a feed-forward 4DGS backbone comparable in scale to Objaverse-XL.
 - *What the next paper should look like*: a transformer-based densifier that consumes monocular video at 30 FPS and emits a 4DGS scene in one pass, evaluated on DyCheck [24].
 - *Mobile outlook*: feed-forward inference maps directly to the 30 FPS sustained target expected on current Snapdragon-class mobile GPUs.
 - *Quantitative target*: by 2027 H1, a feed-forward 4DGS backbone that reconstructs a 10-second DyCheck clip in <30 s on a single mobile GPU, matching the per-scene optimisation quality of Deformable-3DGS (LPIPS ≤ 0.18) within 5%.
- (2) **Hardware-software co-design.** Neo [21]’s Reuse-and-Update sorting accelerator hints at dedicated 4DGS compute units inside mobile GPUs.
 - *Missing piece*: a public 4DGS workload characterisation (FLOPS, tile occupancy, register pressure) on Adreno-class mobile GPUs.
 - *What the next paper should look like*: a tile-sort accelerator IP synthesised for a mobile shader core, benchmarked against the Vulkan 1.3 compute path.
 - *Mobile outlook*: the Adreno shader ISA and the Vulkan 1.3 compute-shader model are the candidate substrate for a co-designed sort accelerator.
 - *Quantitative target*: a co-designed sort accelerator that reduces the per-Gaussian sort cost by 3× versus the software-only Vulkan 1.3 baseline, measured on the Mip-NeRF 360 bicycle scene at 1080p.
- (3) **4DGS ray tracing.** The 4DGRT [51] route is the long-term path to high fidelity, but mobile ray tracing remains an open problem.
 - *Missing piece*: mobile RT cores are bandwidth-starved for the per-Gaussian intersection workload; no published 4DGS BVH layout.
 - *What the next paper should look like*: a two-level BVH that exploits Gaussian covariance locality, evaluated against the rasteriser baseline on a Snapdragon-class mobile GPU.
 - *Mobile outlook*: mobile RT is not on the immediate Vulkan 1.3 hot path; keep as a 2027+ watch item.
 - *Quantitative target*: by 2028, a BVH-based 4DGS ray tracer that reaches 15 FPS @ 720p on a future Snapdragon (Gen 5 or later) mobile GPU, within 2× of the rasteriser FPS. (This is a survey-side projection, not a 4DGRT [51] measurement; 4DGRT reports only desktop RTX 4090 / OptiX numbers, e.g. 36.56 FPS pinhole / 3.44 FPS fisheye, and does not project to mobile hardware.)
- (4) **Temporal compression.** Dynamic / static separation + residual coding + contextual coding may converge into a 4DGS compression standard.
 - *Missing piece*: an MPEG-style codec profile that standardises the dynamic / static split, the residual format, and the rate-control interface.
 - *What the next paper should look like*: a reference encoder / decoder pair with bitstream conformance tests, integrating 4DGS-CC [68] and P-4DGS [28].
 - *Mobile outlook*: a codec standard directly addresses the storage bottleneck observed in the surveyed mobile deployments.
 - *Quantitative target*: a reference codec that compresses the 4DGS-1K D-NeRF scenes to <20 MB per scene at PSNR ≥ 30 dB, with a decoder-side decode time under 50 ms on a current mobile GPU.
- (5) **Generative 4DGS + cross-domain fusion.** The L2D2-GS [61] feed-forward route (§4) generalises to a wider class of generative 4DGS models. The natural extension is text-to-4DGS (analogous to DreamFusion’s text-to-3D path) and image-to-4DGS (single-image dynamic scene reconstruction). Cross-domain fusion closes the loop between 4DGS reconstruction and adjacent modalities (GaussianFluent [16] for physical simulation, 4DGS-SLAM for tracking, and on-device large multimodal models for grounding).
 - *Missing piece*: a public, large-scale text-paired dynamic video corpus (comparable in scale to Objaverse-XL or

LAION-5B for static 3D generation) and an end-to-end system paper that closes the loop between 4DGS reconstruction, physics simulation, and on-device LLM/VLMget at 60 FPS, see §5) is a structural artefact of the rasteriser’s α -blend rule (Equation (3)). No algorithm change can amortise that cost; only a hardware change can. Neo’s Reuse-and-Update sort (§5) is a software prototype of what a dedicated sort accelerator would do, and the 3–5 $\times$ sort speed-up it delivers would translate directly to a 30–50% end-to-end speed-up at the 30 FPS sustained target.

- *What the next paper should look like:* a diffusion-based 4DGS generator that consumes a text prompt + a single static 3DGS seed and outputs a 4DGS scene in <10 seconds on a current mobile GPU, evaluated on a held-out subset of DyCheck prompts; OR a “video in, simulator-in-the-loop 4DGS out” demo running in real time on a mobile device.
 - *Mobile outlook:* exciting but currently compute-bound; a mobile target is plausible only after the mobile transformer cost analysis matures.
 - *Quantitative target:* by 2027 H2, a text-to-4DGS system that produces a 10-second dynamic clip on a single mobile GPU in <30 seconds, with LPIPS ≤ 0.20 against the ground-truth clip; OR a cross-domain demo that combines 4DGS reconstruction with a fracture-aware simulation and renders the result at ≥ 10 FPS on a mobile device.
- (6) **4DGS standardisation.** The 4DGS community currently has no standard codec profile, no standard benchmark suite, and no standard mobile evaluation harness. Three pieces of standardisation are likely in the next two years.
- *Missing piece:* an MPEG- or Khronos-style codec profile that standardises the dynamic / static split, the residual format, and the rate-control interface. The 4DGS-CC [68] + P-4DGS [28] + PD-4DGS trio are the de-facto building blocks.
 - *What the next paper should look like:* a reference encoder / decoder pair with bitstream conformance tests, distributed as an open-source reference implementation.
 - *Mobile outlook:* the reference codec is the natural first deliverable for any mobile-4DGS working group.
 - *Quantitative target:* a reference decoder that runs at 30 FPS sustained on a Snapdragon-class mobile device, on a 20 MB 4DGS bitstream, with PSNR ≥ 30 dB.

8.2 Why Hardware–Software Co-Design Will Dominate

We expect *hardware–software co-design* to matter more than algorithmic progress for mobile 4DGS over the next two years. The argument has three legs.

First, the algorithmic headroom is shrinking. The 4DGS literature has seen roughly 3 $\times$ per-year gains in training speed (4DGS 30–60 min in 2023, OMG4 22 min in 2025, L2D2-GS <1 s in 2026 at inference) and 5–10 $\times$ per-year gains in model size (vanilla 4DGS 60–200 MB in 2023, 4DGS-1K ~7–50 MB (D-NeRF aggregate) / ~50–420 MB (N3V aggregate) in 2025, with per-scene ranges D-NeRF 3.56–16.11 MB and N3V 29.34–69.24 MB). Both curves are flattening: the next 3 $\times$ in either dimension will require a fundamentally new representation (e.g. generative 4DGS, see §8.5 above), not an incremental tweak of the existing one.

Second, the hardware headroom is large and quantifiable. The Adreno-class mobile GPU’s tile-sort cost (10–20% of the frame budget at 60 FPS, see §5) is a structural artefact of the rasteriser’s α -blend rule (Equation (3)). No algorithm change can amortise that cost; only a hardware change can. Neo’s Reuse-and-Update sort (§5) is a software prototype of what a dedicated sort accelerator would do, and the 3–5 $\times$ sort speed-up it delivers would translate directly to a 30–50% end-to-end speed-up at the 30 FPS sustained target.

Third, the co-design loop is short. The Snapdragon mobile launch cadence is roughly 12 months (Gen 3 $\rightarrow$ Gen 4 $\rightarrow$ Gen 5 in 2024/2025/2026), which means the algorithm-to-silicon loop is at most 24 months. The Adreno shader ISA is stable across generations (no breaking changes from Gen 3 to Gen 4), and the Vulkan 1.3 driver is upstreamed into the Linux kernel with a 6-month cadence. These three properties together make a tight algorithm / driver / ISA co-design loop tractable, which is the structural advantage that mobile 4DGS has over its desktop counterpart. The contrarian view is therefore not “algorithm doesn’t matter” but rather “the next 3 $\times$ on mobile will come from hardware / driver / ISA co-design, not from a better MLP”.

8.3 Broader Outlook

The five directions above are not project-specific; they generalise to any mobile-SoC target that combines a tile-based deferred rasteriser with a Vulkan 1.3+ driver (Qualcomm Snapdragon 7/8 series, MediaTek Dimensity 9000+, Samsung Exynos 2400, etc.). The hardware-software co-design thesis is a community-wide observation, not tied to any single application. The Section 6 quantitative anchors (Flux-GS, Mobile-GS, GS-NFS, Lumina) serve as the on-device measurement floor against which the directions above should be evaluated.

8.3.1 Additional related work surveyed for completeness. Beyond the 60 most-cited works discussed above, the present survey also covers nine additional papers that inform the future-directions discussion but did not fit naturally into the in-line citations. They are cluster-cited here across four thematic axes: **compression and dynamic/static separation** (CATSplat [2], MaGS [6], LUDVIG [5], SplatTalk [8], BezierGS [1]); **self-calibrating and self-supervised reconstruction** (Self-Cali-GS [7], GaussianOcc [4]); **single-image free-pose generation** (FreeSplatter [3]). Each entry serves as a borderline-or-out-of-faction reference that expands the survey’s coverage without diluting the four core research directions.

9 CONCLUSION

We have systematised more than 180 representative works on real-time dynamic 4DGS rendering on mobile GPUs published between 2023 and the first half of 2026. The survey is organised around four themes: representation, training-time acceleration, rendering-time acceleration, and mobile deployment. Current Snapdragon 8 Elite devices already achieve desktop-class real-time 3DGS, but 4DGS remains limited by a triple bottleneck of memory bandwidth, Vulkan driver maturity, and the mobile power wall.

Promising future directions include feed-forward 4DGS, hardware–software co-design, 4DGS ray tracing, a temporal compression standard, and cross-modal fusion. We expect the emergence of a community-wide *4DGS compression standard*—a codec profile that combines dynamic/static separation, residual coding, and rate–distortion optimisation; preliminary prototypes along this line include 4DGS-CC [68] and the feed-forward densification policy of L2D2-GS [61], both of which could anchor such a standard. This survey is the academic counterpart of a separate engineering roadmap (the authors’ production-track 4DGS renderer); the survey emphasises breadth across the research landscape rather than the engineering decisions of any one project.

9.1 Limitations of This Survey

We acknowledge three known limitations of the present survey. First, in **scope**: we cover 2023 to the first half of 2026; very early 4DGS works from 2022 (e.g. the original NeRF-t lineage) and any submission that appears after the cut-off are excluded. Second, in **methodology**: the survey is English-only and prioritises code-available or reproducible papers; non-English venues and theoretical-only submissions are underrepresented. Third, in **known gaps**: (i) Faction A (the earliest 4DGS canonical-space methods) receives less space than the newer feed-forward and compression factions; (ii) feed-forward 4DGS is covered via L2D2-GS but the parallel diffusion-based generation literature is only sampled; and (iii) cross-domain fusion papers (4DGS + SLAM, 4DGS + simulation) are sparse and we surface them only in §8.

9.2 Open Science and Reading Notes

In the spirit of survey reproducibility, the authors maintain detailed per-paper notes (not included in this submission) that record the source PDF page for every quantitative claim, the citation chain, and a faction-classified summary. Every \cite key in this PDF corresponds to a paper note in this private collection, which the authors are happy to share on request. Readers who spot missing papers, mis-categorised factions, or updated performance numbers are invited to contact the corresponding author.

Section reading notes: the per-paper notes collection.

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The citekey in the previous draft was wrong: sec-7 used [43] for the 12-camera Ballroom description, but li2022plenovdb is the 15-18-camera N3V/Neural 3D Video (6 indoor scenes). v5.45 fix: split into two cite keys, li2022plenovdb for N3V and li2022plenopticvideostream for Plenoptic Video Dataset.. [43] Tianye Li, Mira Slavcheva, Michael Zollhoefer, Simon Green, Christoph Lassner, Changil Kim, Tanner Schmidt, Steven Lovegrove, Michael Goesele, Richard Newcombe, Zhaoyang Lv. 2022. Neural 3D Video Synthesis from Multi-view Video. paper-notes/2022-li-plenoptic-video.md. NOTE: this is the Neu3D / Neural 3D Video Synthesis paper (arxiv 2103.02597), commonly cited in 4DGS literature; the citekey "plenovdb" is a portmanteau of "Plenoptic" + "VDB" but the paper title is "Neural 3D Video Synthesis" —not "Plenoptic Video: A Simple Pipeline for Volumetric Video Streaming" (a different paper, sometimes confused with Neu3D).. [44] Waseem Mousa, Alaa Maalouf. 2026. 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